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**Lab Work Report**

Introduction to Mobile App Design and Development  
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## **Lab #2: App Navigation and Git**

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## 1 GitHub Repository

Due to the nature of having multiple apps, I created two different GitHub repos since I already had one repo created for the last assignment, I will continue to use that for both the reports and one app, while the other houses the other git demo app. The link for the first is: <https://github.com/konradszulc/konradszulc.introMobileLabs>. This first link contains two branches, one being the master (default) which contains the app files with the navigation side menu, the second branch report is for each lab report created for myself to keep track but is also viewable. The link for the second is: <https://github.com/konradszulc/GitDemo>. This second link contains the demo blank app for testing of creating and pushing to GitHub with Git.

## 2 Learning Outcome and Reflection

From the start of this module, I already felt it was quite the knowledge check as compared to the previous module. There was quite a bit of complex information of the structure of the code for mobile apps. While I have worked on HTML before, I have limited experience with JavaScript. Therefore, the material provided, and the Ionic Docs was useful for understanding what exactly I was looking at in the syntax. Nonetheless, once I completed the tasks and felt like I understood the content a bit better, it was a rewarding feeling, and I am excited for the other module's new content that I had never experienced before.

Before starting the assignment however, I first went through all the material provided. While at first it was overwhelming, by thoroughly reading and practicing along the various videos it slowly started to make more sense. I do find that Ionic seems to be quite straightforward, and with the Docs available eases the practice of putting in various features on a mobile app. I have a bit more experience with GitHub and Git that the videos were a bit more of a reminder rather than new content. Learning more about CLI commands and to directly work with the files on a local machine to be pushed to the Repo is very important and useful. The two videos offered by FreeCodeCamp.org (2020) and Colt Steele (2019) provided valuable insight into why the git commands function as they do, with great live examples and clear instructions to follow along in testing. As well, giving use cases such as trying to push or merge branches and overcoming different errors that may arise, while also explaining why those errors occur is needed for future projects.

Some of the new things I learned were being able to create buttons, and without the need of using the tabs template, creating my own tabs as much as possible with icons provided by Ionics libraries. As well, Simon Grimm's (2022) video provided valuable info, not only to be able to create the side menu navigation, which was new, but also understanding how to create a login page, which I believe will be useful for future modules and the project. I would not say I was surprised at first of the ease of using components, because to me it was quite confusing at first to figure out how to use all those components. However, once I started noticing the pattern such as IonMenu or IonItemSliding and the ease of snippets, it became more

understandable what part of the syntax was required, and I am eager to continue learning what other components to create a beautiful app.

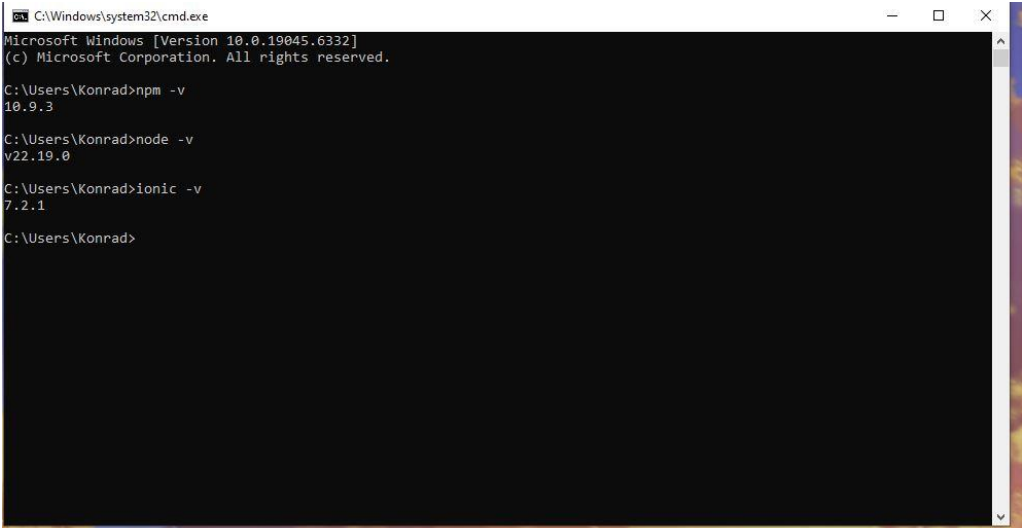
Furthermore, the guide by Kalpana Rathod and Paresh Rathod (2025) was necessary to complete this task, but I did find myself lost to some of the steps. In particular, I was confused why we required some of the lesser-known extensions in VS Code, curious if they were all necessary. Nonetheless, the guide led to a result I am happy with. As well, the Udacity course about mobile constraints and the features of android, are valuable things to constantly consider when designing a UI/UX. To think about the user's journey with the app, will the user be on a one handed or two-handed device, what kind of connection they are on, or responsiveness are crucial to developing a 5-star app. I shall continue to remind myself of the benefits of mobile but also constraints when designing my team project app. Lastly, the most technical heavy material were the typescript video by Mosh Hamedani (2022), which focused on the basics of typescript for beginners such as myself to get a foothold, by describing what it is, but also talking about different types and functionalities with live practice. The react tutorial by Mosh Hamedani (2023) operates in the same way by developing a beginners thought process how to deliver a scalable application with react, providing exercises to follow along with, and describing in great deal many different tasks that may come up for a beginner. I do think it is impossible for a beginner such as myself to understand all these concepts after just a couple of videos, that is why practicing and constant usage of these videos over time, will help me to become more proficient in the developing stages of an app.

I did contact some friends in relation to this assignment, to clarify the instructions and provide clarity on which exact dependencies we should install. I then understood that by having most of them, it provides more flexibility for future projects. As well, our tutor's response in the discussion page of the course, mentioning about separate git repositories was needed, as I did have concern about pushing both apps to one repo. I am glad to have such communication with friends, and our tutor to ease any difficulties in the assignments.

Some of the trouble I encountered during this assignment were minor inconveniences that took a bit of investigating. For example, when working with Visual studio code and the Ionic app, I kept having issues with trying to run Ionic serve. It stated that permissions were not set, I eventually realized it was due to the terminal being in powershell rather than CMD on VS Code. Once that was changed there was no longer an error. Another example I experienced was when testing the app, I created when cloned from the Repo, as I could not direct run ionic serve from the cloned version but the original version. I eventually realized that I had to do npm install for all the local dependencies before I can run a cloned repo of an Ionic app. It was cases like these where StackOverflow provided some useful tips from all the other users. While I have not joined StackOverflow yet, I understand how helpful in the coding community it can be for novices such as myself.

### 3 Confirming Ionic Installation with the proper dependencies

First and foremost, I verified that I have everything required from last module installed. I then proceeded to install all dependencies mentioned on the guide. However, I noticed that even though I successfully installed Ionic and followed every instruction, there was no specific way to install Cordova or native run from the task instructions. Thus, I utilized the Ionic Docs to find the right commands such as `npm install -g @ionic/cli@latest native-run` and `npm install -g @ionic/cli@latest cordova-res`. Which ultimately got my ionic packages up to date and ready for my future projects. This was not a difficult task, as the Ionic docs was quite informative at every step, and everything was smoothly installed.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

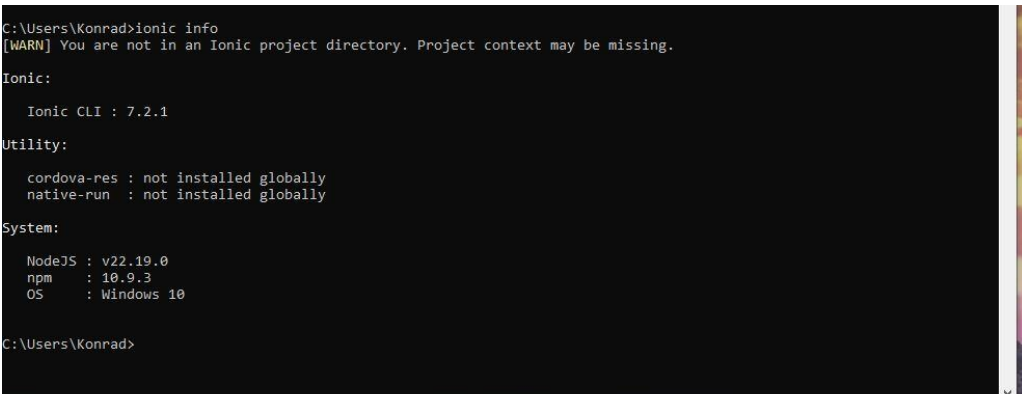
C:\Users\Konrad>npm -v
10.9.3

C:\Users\Konrad>node -v
v22.19.0

C:\Users\Konrad>ionic -v
7.2.1

C:\Users\Konrad>
```

Figure 1 Confirming Installation of npm/node/ionic



```
C:\Users\Konrad>ionic info
[WARN] You are not in an Ionic project directory. Project context may be missing.

Ionic:

  Ionic CLI : 7.2.1

Utility:

  cordova-res : not installed globally
  native-run  : not installed globally

System:

  NodeJS : v22.19.0
  npm    : 10.9.3
  OS     : Windows 10

C:\Users\Konrad>
```

Figure 2 Noticed no Cordova or Native-run installed

```
C:\Users\Konrad>npm install @ionic/angular@latest --save -g
added 14 packages in 7s
```

Figure 3 Installed Angular

```
C:\Windows\system32\cmd.exe
C:\Users\Konrad>npm install @ionic/react -g
added 9 packages in 5s
C:\Users\Konrad>npm install @ionic/react-router -g
added 26 packages in 6s
C:\Users\Konrad>
```

Figure 4 Installed React

```
C:\Users\Konrad>ionic info
[WARN] You are not in an Ionic project directory. Project context may be missing.

Ionic:

  Ionic CLI : 7.2.1

Utility:

  cordova-res : 0.15.4
  native-run  : 2.0.1

System:

  NodeJS : v22.19.0
  npm    : 10.9.3
  OS     : Windows 10

C:\Users\Konrad>
```

Figure 5 Ionic with all dependencies

#### 4 Installing all necessary plug-ins for Visual Studio Code

While I installed everything mentioned by the guide, even the optional ones, I was concerned by some of the lower download extensions and random publishers. However, I proceeded to give trust to these extensions as the ones that were not from a verified creator were at least simple, or a bundle of other verified creators such as the expansion pack. VS Code's extension plug in installer is quite straightforward, which as seen in figure 6 all extensions in one place.

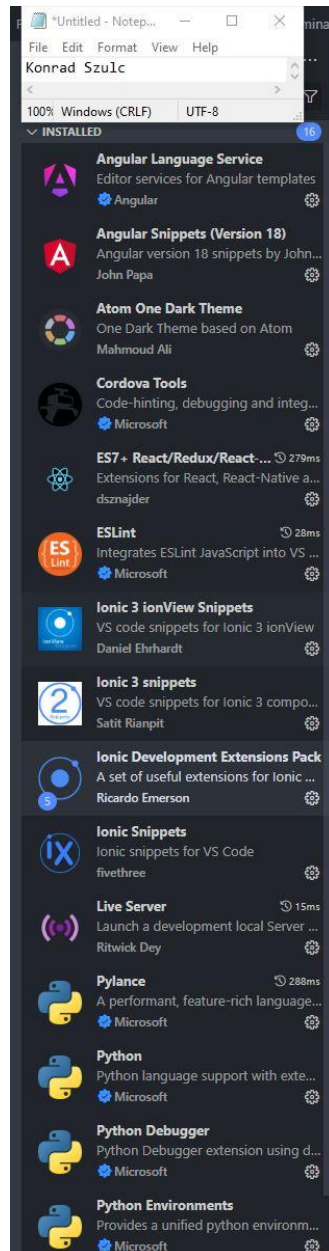


Figure 6 All recommended plug ins up to date in VS Code

## 5 Creating the Ionic App, and coding the Navigation side menu bar

This to me was the quite harder part of the two tasks, as I mentioned earlier in the learning reflection, seeing such new code structures can be overwhelming. Nonetheless, by breaking it down step by step and utilizing the guide/videos, I was able to accomplish the task with little friction. Since I have already created an app before in module 1, the steps to do so again were not difficult. I had not ported over the code to VS Code, so understanding it as one simple command was quite amusing. Going forward, while I have coded before, understanding where everything is

located was a bit jarring. Having a guide to explain the main files and pages was very useful. Comparing it to HTML, there are a lot of similarities, thus while at first it is confusing, it is essentially just a new similar language to create a feature you want. The figures go over the exact things I changed while also some being highlighted to illustrate small code changes. I was quite proud to have a simple app with navigation already fully functional.

```

C:\Users\Konrad\Desktop\Labwork-2
C:\Users\Konrad\Desktop\Labwork-2>ionic start --list
Starters for @ionic/vue (--type=vue)

name | description
-----|-----
tabs | A starting project with a simple tabbed interface
sidemenu | A starting project with a side menu with navigation in the content area
blank | A blank starter project
list | A starting project with a list

Starters for @ionic/angular (--type=angular)

name | description
-----|-----
tabs | A starting project with a simple tabbed interface
sidemenu | A starting project with a side menu with navigation in the content area
blank | A blank starter project
list | A starting project with a list
my-first-app | A template for the "Build Your First App" tutorial

Starters for angular-standalone (--type=angular-standalone)

name | description
-----|-----
tabs | A starting project with a simple tabbed interface
sidemenu | A starting project with a side menu with navigation in the content area
blank | A blank starter project
list | A starting project with a list
my-first-app | A template for the "Build Your First App" tutorial

Starters for @ionic/react (--type=react)

name | description
-----|-----
blank | A blank starter project
list | A starting project with a list
my-first-app | A template for the "Build Your First App" tutorial
sidemenu | A starting project with a side menu with navigation in the content area
tabs | A starting project with a simple tabbed interface

```

Figure 7 Used tab template

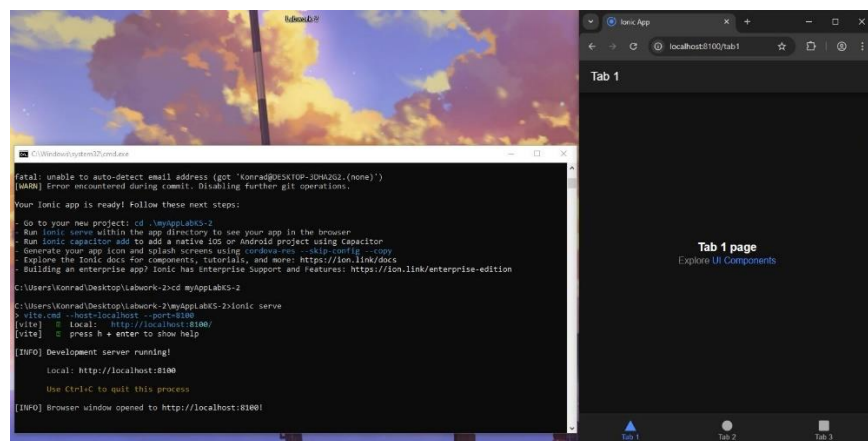


Figure 8 created base tab template app



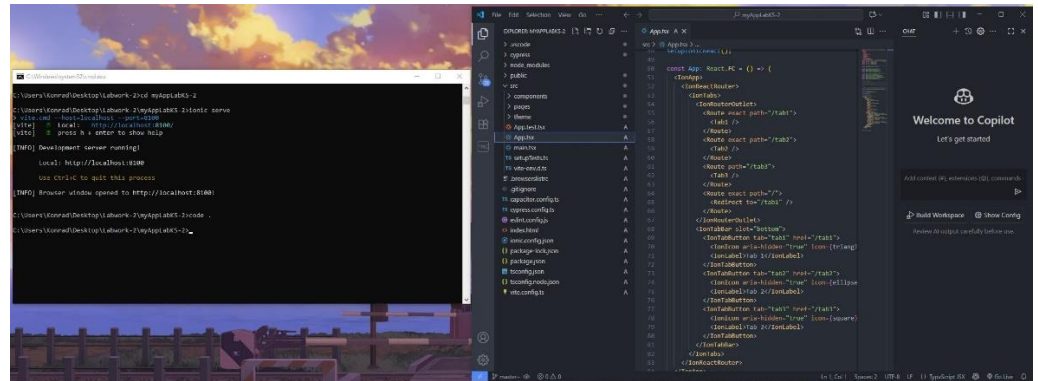


Figure 9 Brought the code over to VS code

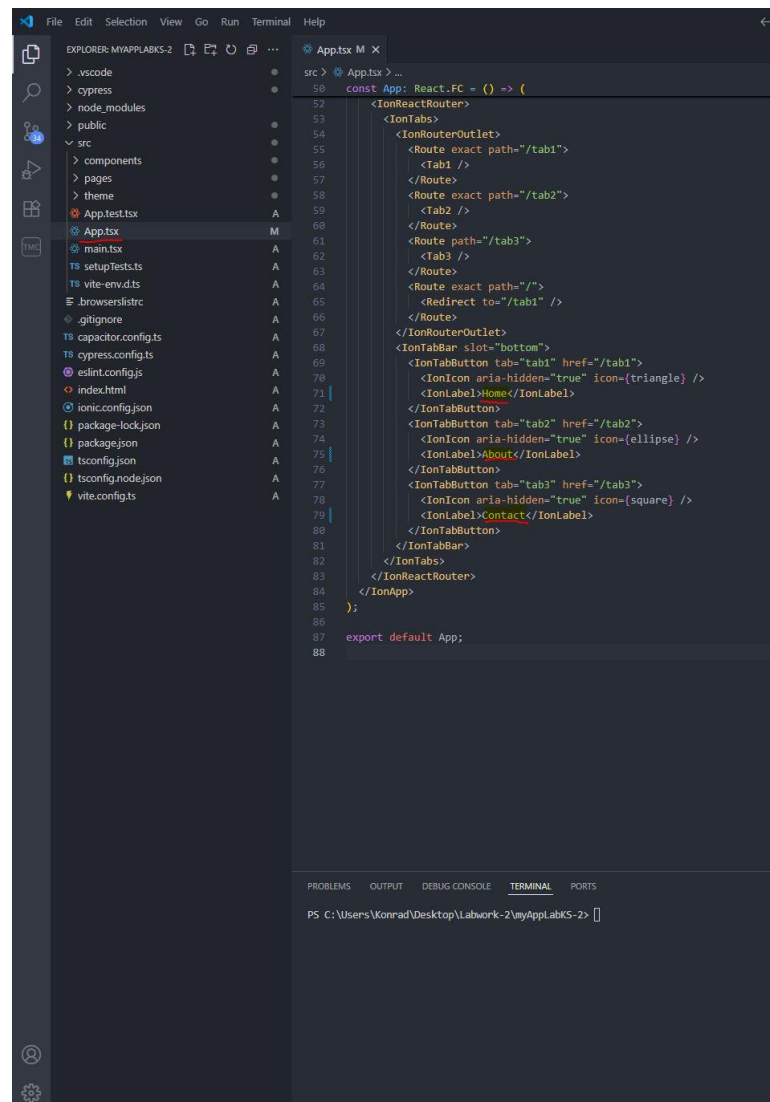
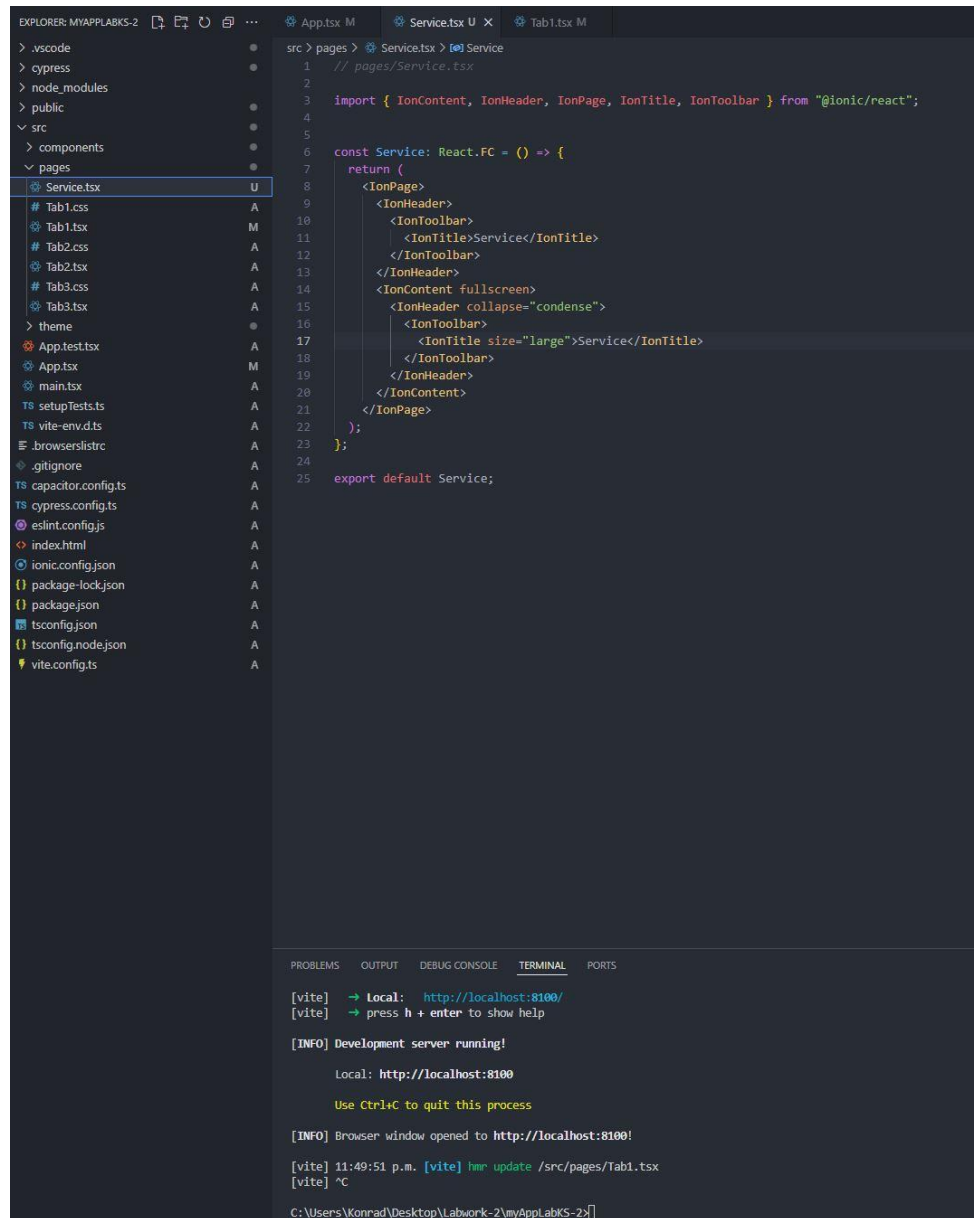


Figure 10 Updated tab names



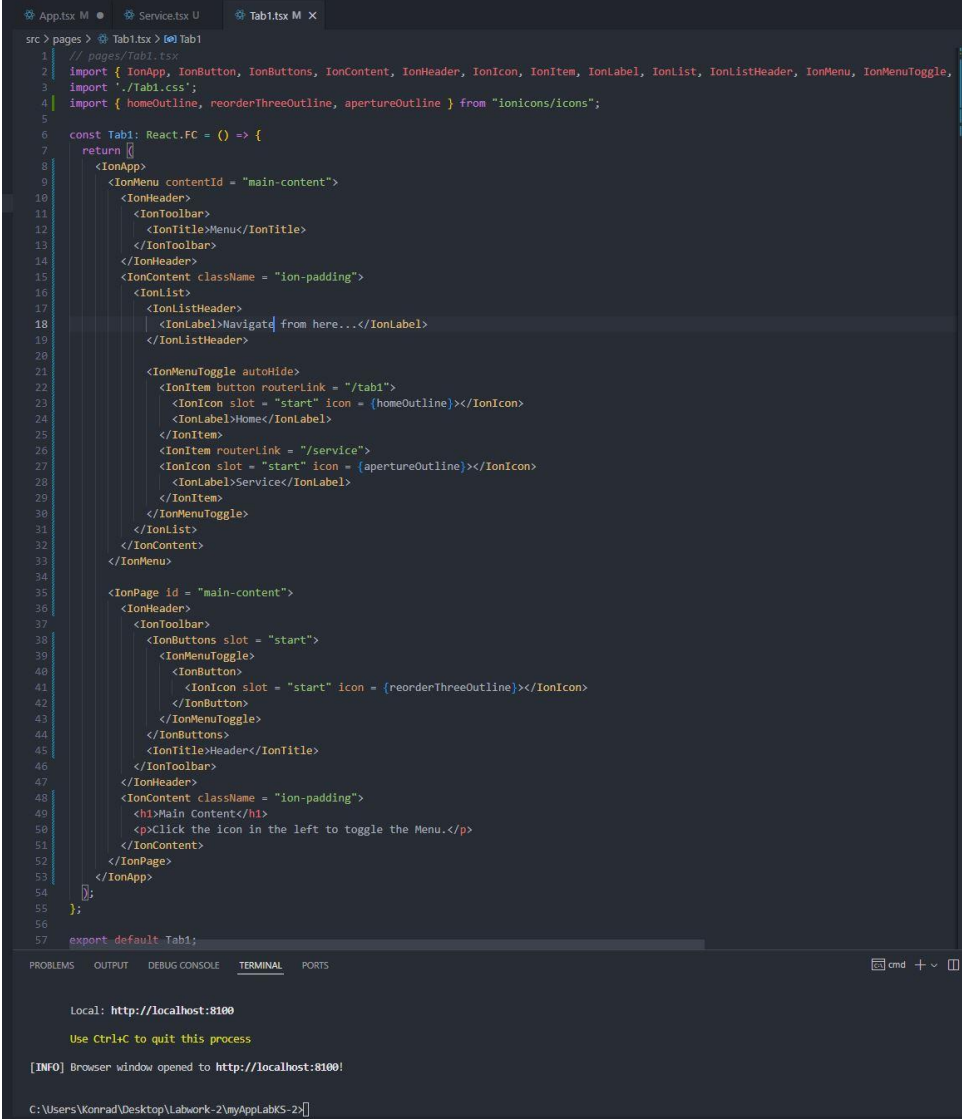
The screenshot shows a VS Code editor with the Explorer sidebar on the left, displaying a file tree for a project named 'MYAPPLABKS-2'. The file tree includes folders like '.vscode', 'cypress', 'node\_modules', 'public', 'src', 'components', and 'pages'. The 'pages' folder is expanded, showing 'Service.tsx' (selected), 'Tab1.tsx', 'Tab2.tsx', and 'Tab3.tsx'. The main editor area displays the content of 'Service.tsx', which is a React component using Ionic React. The code defines a 'Service' component that renders an 'IonPage' with an 'IonHeader' and 'IonContent'. The 'IonHeader' contains an 'IonToolbar' with an 'IonTitle' set to 'Service'. The 'IonContent' is set to 'fullscreen' and contains an 'IonHeader' with 'collapse="condense"' and an 'IonToolbar' with an 'IonTitle' set to 'Service'. The component is exported as the default export.

```
src > pages > Service.tsx | Service
1 // pages/Service.tsx
2
3 import { IonContent, IonHeader, IonPage, IonTitle, IonToolbar } from "@ionic/react";
4
5
6 const Service: React.FC = () => {
7   return (
8     <IonPage>
9       <IonHeader>
10        <IonToolbar>
11          <IonTitle>Service</IonTitle>
12        </IonToolbar>
13      </IonHeader>
14      <IonContent fullscreen>
15        <IonHeader collapse="condense">
16          <IonToolbar>
17            <IonTitle size="large">Service</IonTitle>
18          </IonToolbar>
19        </IonHeader>
20      </IonContent>
21    </IonPage>
22  );
23 };
24
25 export default Service;
```

The terminal at the bottom shows the output of the development server. It indicates that the server is running on <http://localhost:8100/> and provides instructions to press 'h + enter' to show help. It also shows the command 'hmr update /src/pages/Tab1.tsx' being executed.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[vite] → Local: http://localhost:8100/
[vite] → press h + enter to show help
[INFO] Development server running!
Local: http://localhost:8100
Use Ctrl+C to quit this process
[INFO] Browser window opened to http://localhost:8100!
[vite] 11:49:51 p.m. [vite] hmr update /src/pages/Tab1.tsx
[vite] ^C
C:\Users\Konrad\Desktop\Labwork-2\myAppLabKS-2>
```

Figure 11 created and added Service page



```
src> pages> Tab1.tsx Tab1
1 // pages/Tab1.tsx
2 import { IonApp, IonButton, IonButtons, IonContent, IonHeader, IonIcon, IonItem, IonLabel, IonList, IonListHeader, IonMenu, IonMenuToggle,
3 import './Tab1.css';
4 import { homeOutline, reorderThreeOutline, apertureOutline } from "ionicons/icons";
5
6 const Tab1: React.FC = () => {
7   return (
8     <IonApp>
9       <IonMenu contentId = "main-content">
10         <IonHeader>
11           <IonToolbar>
12             <IonTitle>Menu</IonTitle>
13           </IonToolbar>
14         </IonHeader>
15         <IonContent className = "ion-padding">
16           <IonList>
17             <IonListHeader>
18               <IonLabel>Navigate from here...</IonLabel>
19             </IonListHeader>
20
21             <IonMenuToggle autoHide>
22               <IonItem button routerLink = "/tab1">
23                 <IonIcon slot = "start" icon = {homeOutline}></IonIcon>
24                 <IonLabel>Home</IonLabel>
25               </IonItem>
26               <IonItem routerLink = "/service">
27                 <IonIcon slot = "start" icon = {apertureOutline}></IonIcon>
28                 <IonLabel>Service</IonLabel>
29               </IonItem>
30             </IonMenuToggle>
31           </IonList>
32         </IonContent>
33       </IonMenu>
34
35       <IonPage id = "main-content">
36         <IonHeader>
37           <IonToolbar>
38             <IonButtons slot = "start">
39               <IonMenuToggle>
40                 <IonButton>
41                   <IonIcon slot = "start" icon = {reorderThreeOutline}></IonIcon>
42                 </IonButton>
43               </IonMenuToggle>
44             </IonButtons>
45             <IonTitle>Header</IonTitle>
46           </IonToolbar>
47         </IonHeader>
48         <IonContent className = "ion-padding">
49           <h1>Main Content</h1>
50           <p>Click the icon in the left to toggle the Menu.</p>
51         </IonContent>
52       </IonPage>
53     </IonApp>
54   );
55 }
56
57 export default Tab1;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

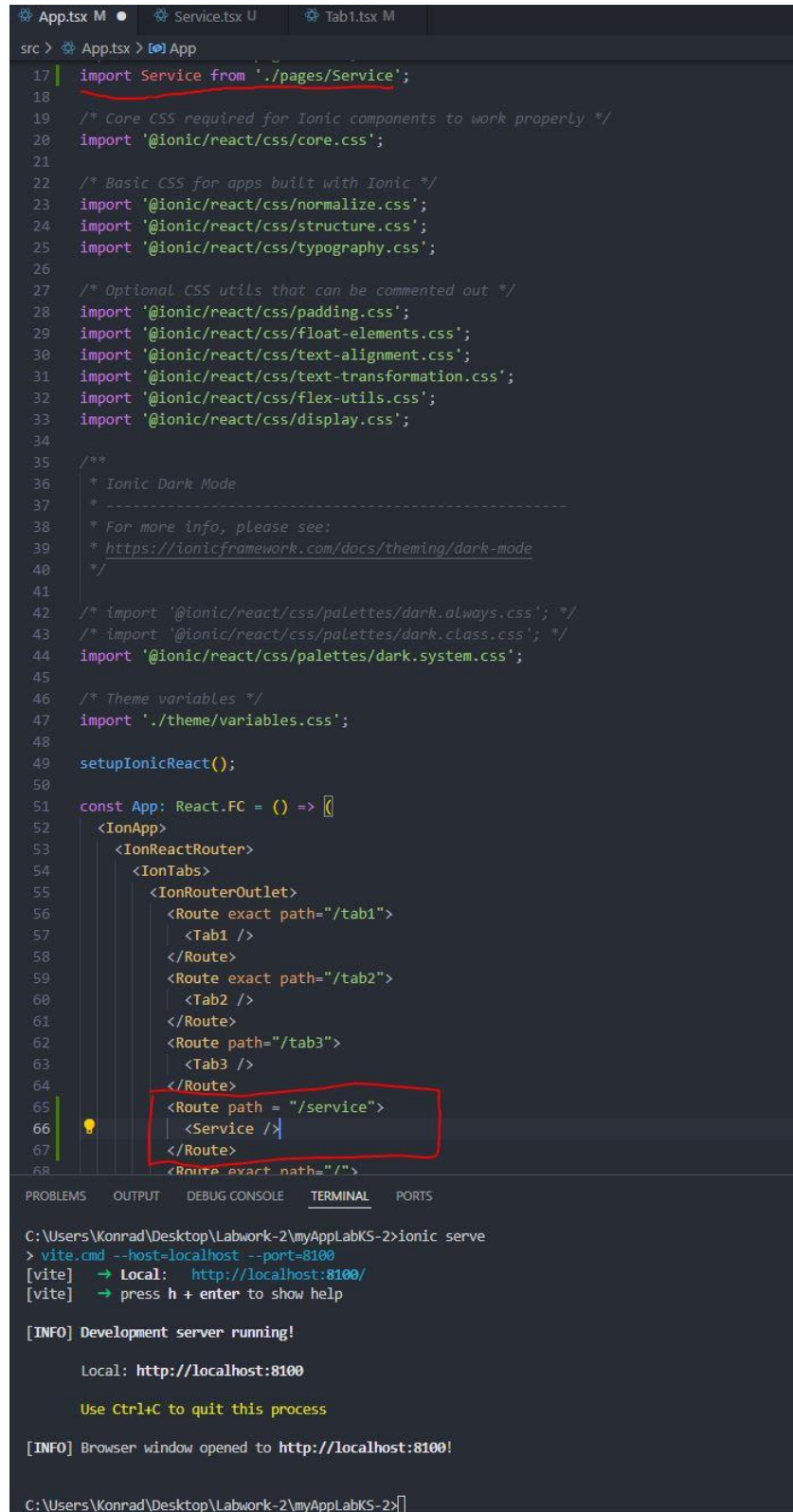
Local: http://localhost:8100

Use Ctrl+C to quit this process

[INFO] Browser window opened to http://localhost:8100/

C:\Users\Konrad\Desktop\Labwork-2\myApp\lab5-2>

Figure 12 created side menu code in Tab 1 page



```
src > App.tsx > App
17 import Service from './pages/Service';
18
19 /* Core CSS required for Ionic components to work properly */
20 import '@ionic/react/css/core.css';
21
22 /* Basic CSS for apps built with Ionic */
23 import '@ionic/react/css/normalize.css';
24 import '@ionic/react/css/structure.css';
25 import '@ionic/react/css/typography.css';
26
27 /* Optional CSS utils that can be commented out */
28 import '@ionic/react/css/padding.css';
29 import '@ionic/react/css/float-elements.css';
30 import '@ionic/react/css/text-alignment.css';
31 import '@ionic/react/css/text-transformation.css';
32 import '@ionic/react/css/flex-utils.css';
33 import '@ionic/react/css/display.css';
34
35 /**
36  * Ionic Dark Mode
37  * -----
38  * For more info, please see:
39  * https://ionicframework.com/docs/theming/dark-mode
40  */
41
42 /* import '@ionic/react/css/palettes/dark.always.css'; */
43 /* import '@ionic/react/css/palettes/dark.class.css'; */
44 import '@ionic/react/css/palettes/dark.system.css';
45
46 /* Theme variables */
47 import './theme/variables.css';
48
49 setupIonicReact();
50
51 const App: React.FC = () => {
52   <IonApp>
53     <IonReactRouter>
54       <IonTabs>
55         <IonRouterOutlet>
56           <Route exact path="/tab1">
57             <Tab1 />
58           </Route>
59           <Route exact path="/tab2">
60             <Tab2 />
61           </Route>
62           <Route path="/tab3">
63             <Tab3 />
64           </Route>
65           <Route path = "/service">
66             <Service />
67           </Route>
68           <Route exact path="/">
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
C:\Users\Konrad\Desktop\Labwork-2\myAppLabKS-2>ionic serve
> vite.cmd --host=localhost --port=8100
[vite] → Local: http://localhost:8100/
[vite] → press h + enter to show help

[INFO] Development server running!

Local: http://localhost:8100

Use Ctrl+C to quit this process

[INFO] Browser window opened to http://localhost:8100!

C:\Users\Konrad\Desktop\Labwork-2\myAppLabKS-2>
```

Figure 13 Added Service route path to App.tsx

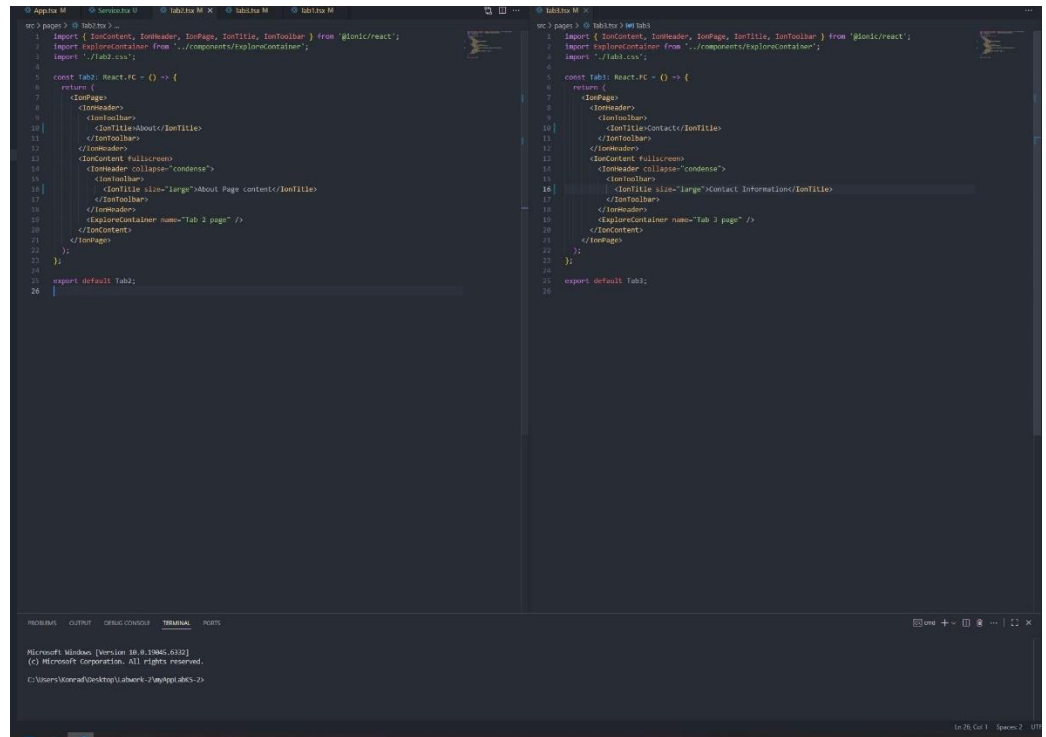


Figure 14 changed the titles for tab 2 and 3

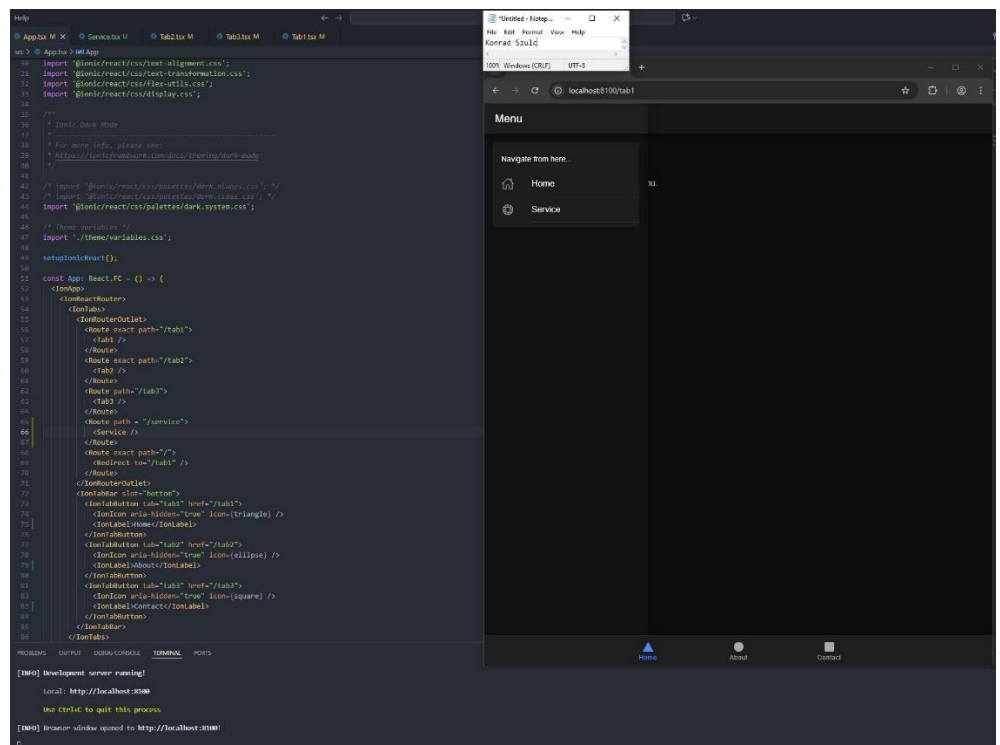
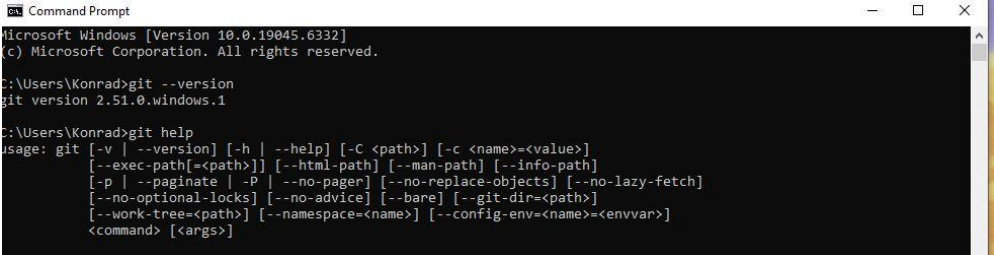


Figure 15 Created the Navigation side menu app

## 6 Working with Git and creating/pushing/pulling/cloning GitHub Repo's

The last task was to understand how to use Git to properly manage changes quickly/smoothly and push the files to GitHub. Like the other task, I confirmed I had everything installed properly and created a new blank tabs app to just test with. Since I already knew the steps of creating another repo, there was no issues at that step. While I have not used the CLI to push the GitHub, usually I dragged and dropped on the GitHub interface to interact with the Repo. I realized quickly this would not work with large amounts of files or be efficient when coding alone or in a team-built application. Thus, learning about these new ways to push the files and interact with Git was useful and beneficial. GitHub also provides similar instructions, making it not too difficult of a task even if it was my first time doing it. Thus, I pushed both apps to separate GitHub repositories and had one repo 2 branches to differentiate the app and the reports. While I did have some errors in regard to trying to push, it was because I had to pay close attention to which branch I was pushing to. Those errors can be seen in figure 21 as I was trying to push it to main, when it was supposed to be master.



```
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Konrad>git --version
git version 2.51.0.windows.1

C:\Users\Konrad>git help
usage: git [-v | --version] [-h | --help] [-C <path>] [-c <name>=<value>]
[--exec-path=<path>] [--html-path] [--man-path] [--info-path]
[-p | --paginate | -P | --no-pager] [--no-replace-objects] [--no-lazy-fetch]
[--no-optional-lock] [--no-advice] [--bare] [--git-dir=<path>]
[--work-tree=<path>] [--namespace=<name>] [--config-env=<name>=<envvar>]
<command> [<args>]
```

Figure 16 confirming Git installation

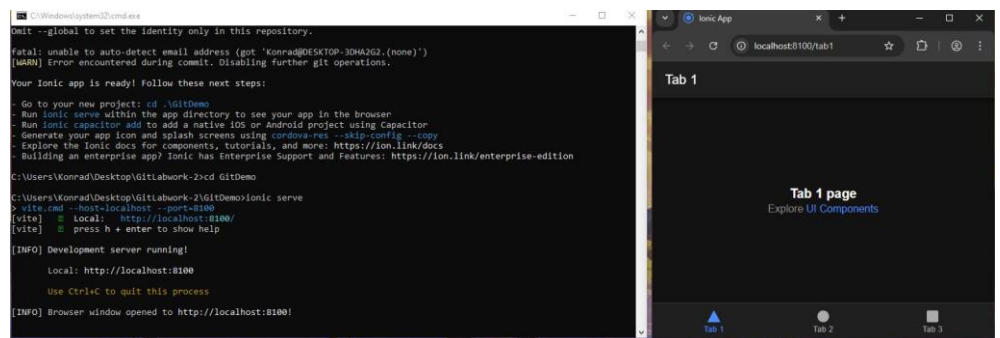


Figure 17 Created New GitDemo App

**Create a new repository**

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).  
Required fields are marked with an asterisk (\*).

**1 General**

**Owner \*** konradszulc / **Repository name \*** GitDemo  
✔ GitDemo is available.

Great repository names are short and memorable. How about **psychic-octo-engine**?

**Description**

0 / 350 characters

**2 Configuration**

**Choose visibility \***  
Choose who can see and commit to this repository. **Public**

**Start with a template**  
Templates pre-configure your repository with files. **No template**

**Add README**  
READMEs can be used as longer descriptions. [About READMEs](#) **Off**

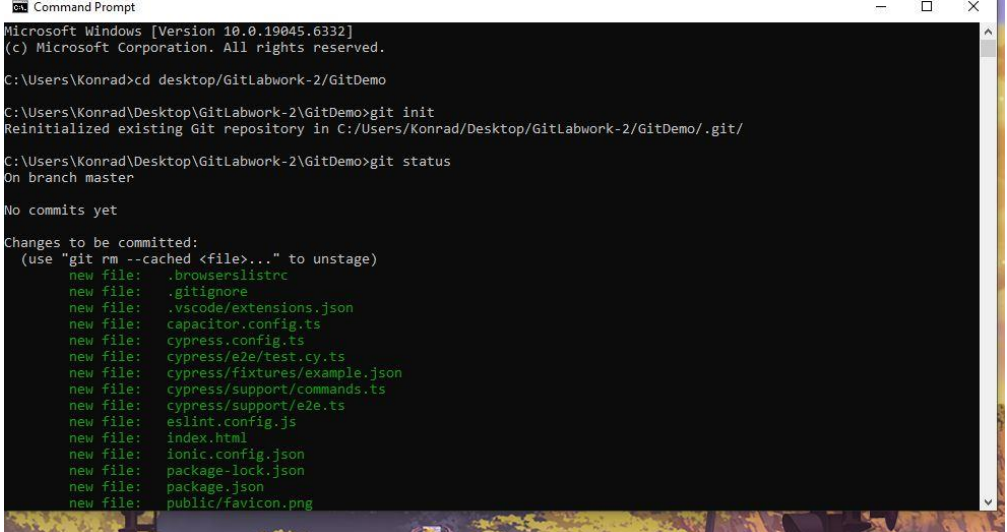
**Add .gitignore**  
.gitignore tells git which files not to track. [About ignoring files](#) **No .gitignore**

**Add license**  
Licenses explain how others can use your code. [About licenses](#) **No license**

**Create repository**

Figure 18 Created Additional GitHub Repo





```
Microsoft Windows [Version 10.0.19045.6332]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Konrad>cd desktop/GitLabwork-2/GitDemo

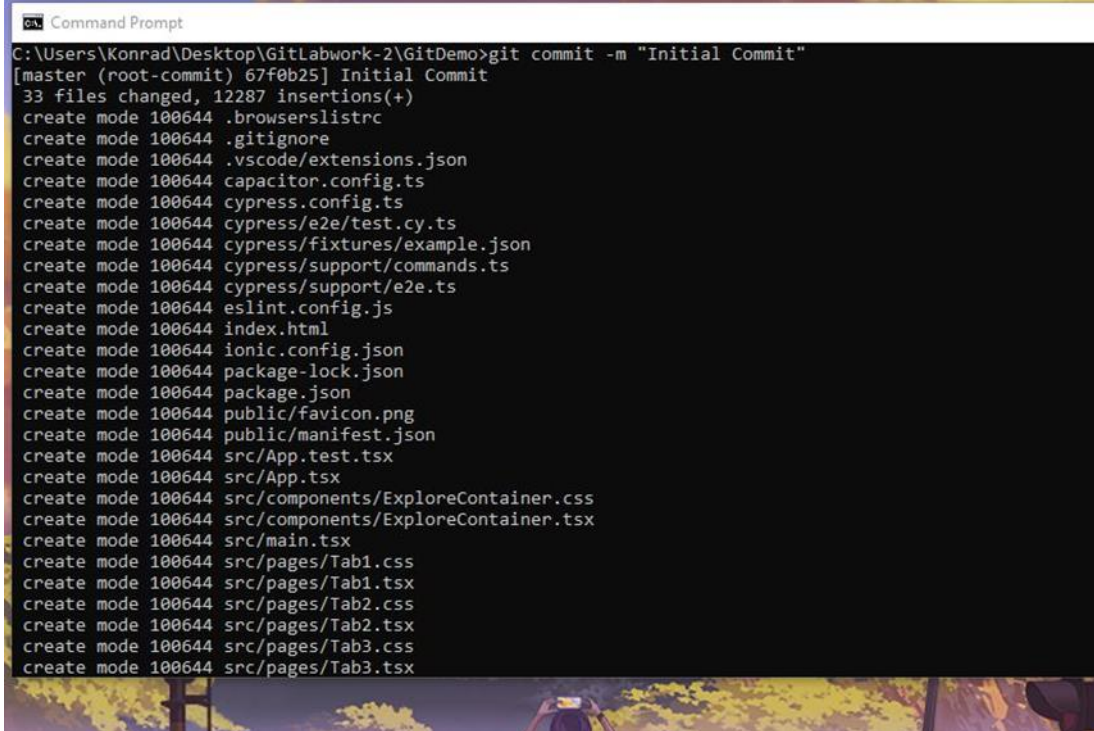
C:\Users\Konrad\Desktop\GitLabwork-2\GitDemo>git init
Reinitialized existing Git repository in C:/Users/Konrad/Desktop/GitLabwork-2/GitDemo/.git/

C:\Users\Konrad\Desktop\GitLabwork-2\GitDemo>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .browserslistrc
        new file:   .gitignore
        new file:   .vscode/extensions.json
        new file:   capacitor.config.ts
        new file:   cypress.config.ts
        new file:   cypress/e2e/test.cy.ts
        new file:   cypress/fixtures/example.json
        new file:   cypress/support/commands.ts
        new file:   cypress/support/e2e.ts
        new file:   eslint.config.js
        new file:   index.html
        new file:   ionic.config.json
        new file:   package-lock.json
        new file:   package.json
        new file:   public/favicon.png
```

Figure 19 Git Init, and Git Status



```
C:\Users\Konrad\Desktop\GitLabwork-2\GitDemo>git commit -m "Initial Commit"
[master (root-commit) 67f0b25] Initial Commit
33 files changed, 12287 insertions(+)
 create mode 100644 .browserslistrc
 create mode 100644 .gitignore
 create mode 100644 .vscode/extensions.json
 create mode 100644 capacitor.config.ts
 create mode 100644 cypress.config.ts
 create mode 100644 cypress/e2e/test.cy.ts
 create mode 100644 cypress/fixtures/example.json
 create mode 100644 cypress/support/commands.ts
 create mode 100644 cypress/support/e2e.ts
 create mode 100644 eslint.config.js
 create mode 100644 index.html
 create mode 100644 ionic.config.json
 create mode 100644 package-lock.json
 create mode 100644 package.json
 create mode 100644 public/favicon.png
 create mode 100644 public/manifest.json
 create mode 100644 src/App.test.tsx
 create mode 100644 src/App.tsx
 create mode 100644 src/components/ExploreContainer.css
 create mode 100644 src/components/ExploreContainer.tsx
 create mode 100644 src/main.tsx
 create mode 100644 src/pages/Tab1.css
 create mode 100644 src/pages/Tab1.tsx
 create mode 100644 src/pages/Tab2.css
 create mode 100644 src/pages/Tab2.tsx
 create mode 100644 src/pages/Tab3.css
 create mode 100644 src/pages/Tab3.tsx
```

Figure 20 Committed the files to Git



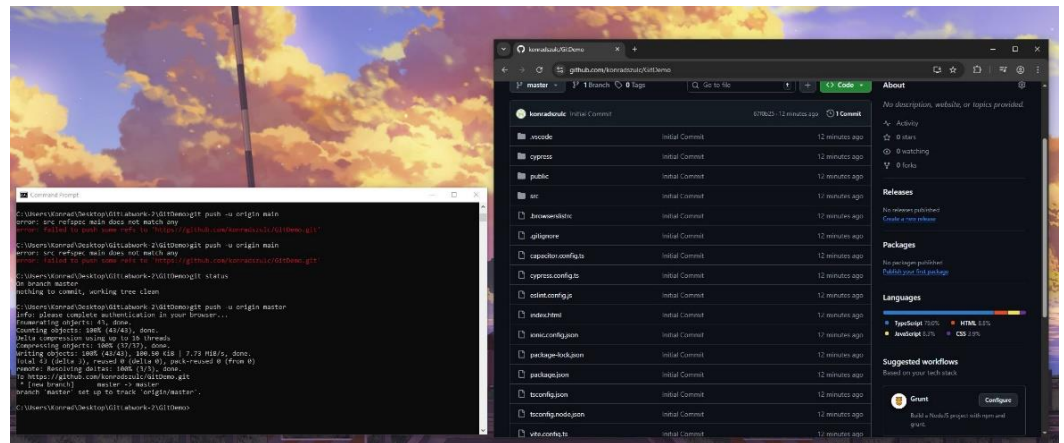


Figure 21 pushed Git demo app to GitHub repo

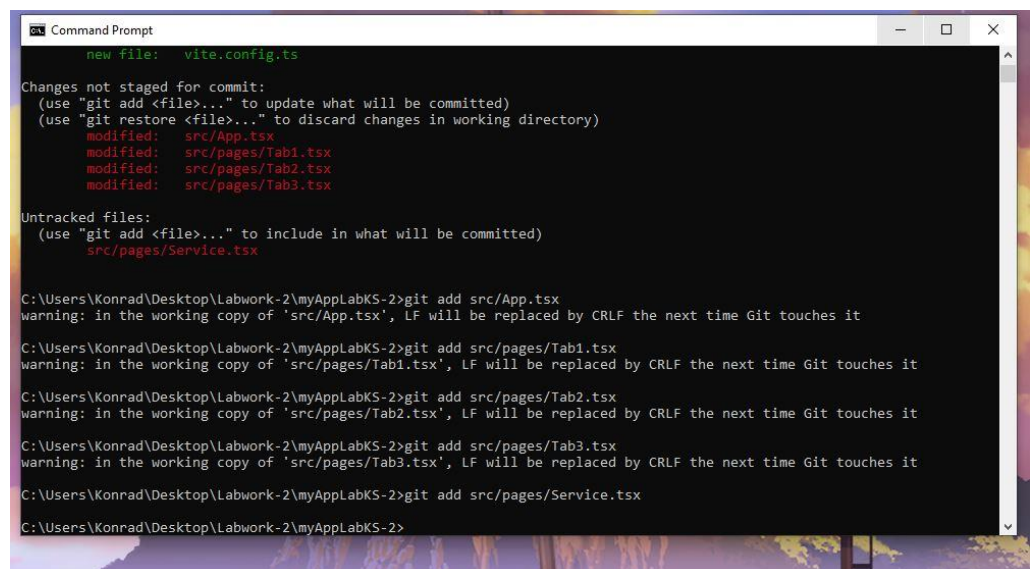


Figure 22 added files to be tracked and staged by Git for Navigation App

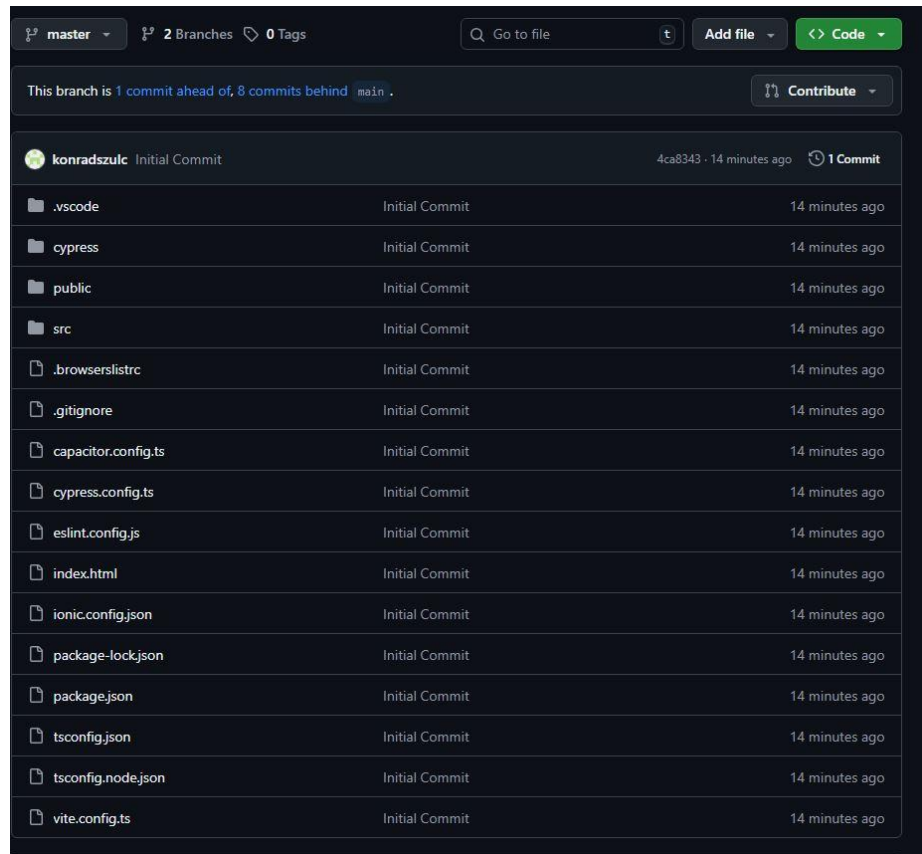


Figure 23 Master branch contains Navigation app code

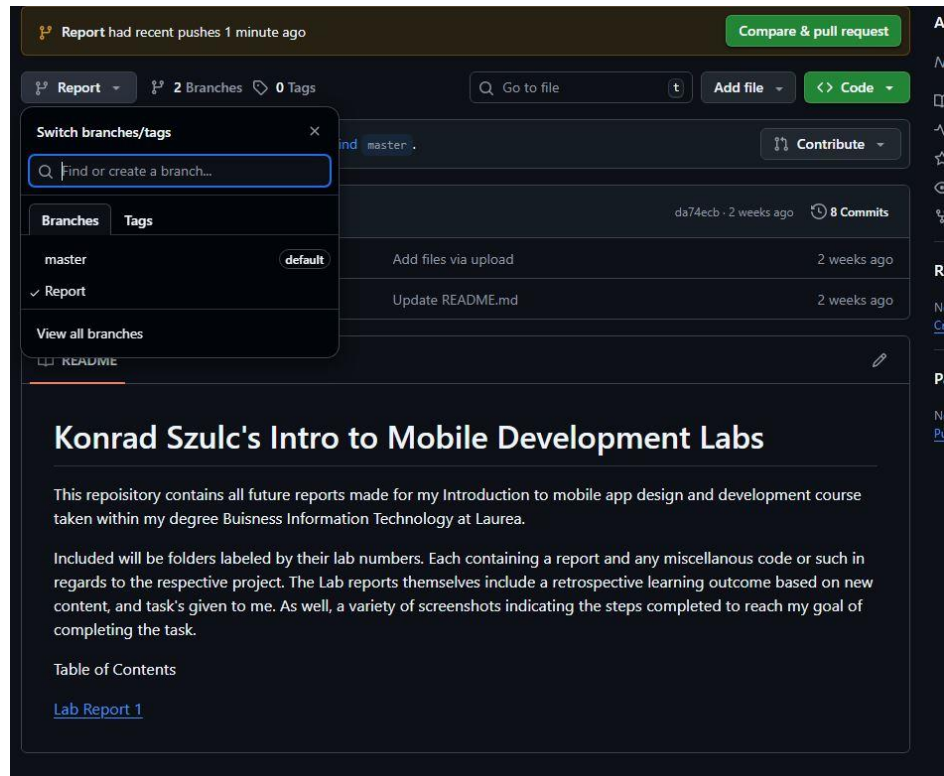


Figure 24 Separate branch for reports

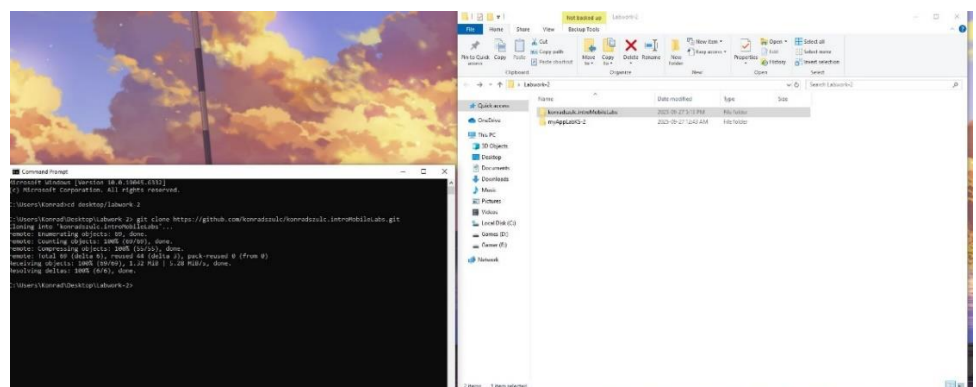


Figure 25 Cloned the Repo for testing purposes

## References

FreeCodeCamp.org. 2020. Git and github for beginners - crash course. YouTube. Accessed on 28 September 2025. <https://www.youtube.com/watch?v=RG0j5yH7evk>

Grimm, S. 2022. Building an ionic react tab bar navigation. YouTube. Accessed on 28 September 2025. <https://www.youtube.com/watch?v=L2qUfEblNj8&t=629s>

Hamedani, M. 2022. Typescript tutorial for beginners. YouTube. Accessed on 28 September 2025. <https://www.youtube.com/watch?v=d56mG7DezGs>

Hamedani, M. 2023. React tutorial for beginners. YouTube. Accessed on 28 September 2025. <https://www.youtube.com/watch?v=SqcY0GIEtPk>

Rathod, K & Rathod, P. 2025. Ionic with git: creating simple tab mobile app with menu navigation. HANDS-ON LABWORK-2 PDF. Accessed 28 September 2025.

Steele, C. 2019. Learn github in 20 minutes. YouTube. Accessed on 28 September 2025. <https://www.youtube.com/watch?v=nhNq2klvi9s>

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